

School of Electrical and Computer Engineering

Department of control system Engineering

Introduction to control system

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Chapter Two
Mathematical modeling of physical system

introduction

- To understand and control complex system, one must obtain quantitative mathematical model of these systems.
- It is necessary to analyze the relationship between the system variables and obtain a mathematical model.
- This model is based upon the fact that the dynamic system can be completely described by known differential equations.
- **Mathematical modeling :** is a set of equations that represents the dynamics of the system accurately or at least fairly well.

- The component a control system are diverse in nature and may include Electrical, Mechanical, Thermal and Hydraulic devices.
- Differential equations describing the dynamic performance of a physical system are obtained by utilizing the **physical laws of the process.**
- Differential equations relating the **input** to the **output** quantities for the control systems are obtained using the basic law of physics .
- The basic law of physics include balancing of force, energy and mass
- For most physical systems are classify the variables as either '**through'** or **'cross'** variables in the sense that
    - **Through refers to a point**
    - **Across refers Between two point**

System	Through variable	Across variable
Electrical	Current, *i*	Potential difference or voltage, V
Mechanical (Translational)	Force ,F	Relative Velocity, V
Mechanical (Rotational)	Torque, T	Relative angular , ω
Thermal	Rate of flow of heat energy, q	Difference in temperature, ΔT
Fluid	Volumetric rate of fluid flow, Q	Difference in pressure, ΔP

Table: A list of analogous variable for different system

Transfer Function

- The transfer function of a linear system is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable, with all initial conditions assumed to be zero.



$$\boldsymbol{G(s) = \frac{C(s)}{R(s)}}$$

- A transfer function may be defined only for a linear, stationary (constant parameter) system. A non stationary system, often called a time-varying system, has one or more time-varying parameters, and the Laplace transformation may not be utilized.

- When a general n^{th}-order, linear, time-invariant differential equations are written

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t)$$

- Where $c(t)$ is the output, $r(t)$ is the input and the a and b and the form of the differential equations represents the system.
- Taking the Laplace transform of both sides, then the solve for the input output ratio.

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0)}$$

- The transfer function of a system (or element) represents the relationship describing the dynamics of the system under consideration.
- Furthermore, a transfer function is an input-output description of the behaviour of a system. Thus, the transfer function description **does not** include any information concerning the **internal structure** of the system and its **behaviour.**

Example

1. Find the transfer function represented by

   a. $\frac{dc(t)}{dt}+2c(t)=r(t)$

   b. $\frac{dc^3(t)}{dt^3}+3\frac{dc^2(t)}{dt^2}+\frac{dc(t)}{dt}+5c(t)=\frac{dr^2(t)}{dt^2}+4\frac{dr(t)}{dt}+3r(t)$

   c. $f(t)=\frac{dx^2(t)}{dt^2}+5\frac{dx(t)}{dt}+10\,x(t)$

2. Find the differential equation corresponding to the transfer function.

$$G(s)=\frac{2s+1}{s^2+6s+1}$$

Advantage

- Transfer function is a mathematical model and it gives the gain of the system.
- Since Laplace transform is used, the terms are simple algebraic expressions and differential terms are not present
- If transfer function of a system is known, the response of the system to any input can be determined very easily.
- Poles and zeros of a system can be determined from the knowledge of the transfer function of the system.

Disadvantage

- Transfer function can be defined for linear systems only.
- Initial conditions lose their importance since transfer function does not take into account the initial condition.
- No inferences can be drawn about the physical structure of a system from its transfer function.

Electrical system

- Newton's laws used to build mechanical systems from components. For electrical systems, we use Kirchhoff's laws to form differential equations for the systems.

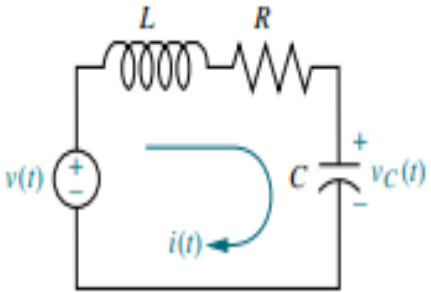
These two laws can be very simply stated:

- **Law 1** – The total current flowing into a node in the circuit is equal to the total current flowing from that node. (i.e. there is no residual current at nodes)
- **Law 2** – In a closed loop of the network, the algebraic sum of the potential differences across each part of the circuit is equal to the applied voltage within that loop

Component	Voltage-current	Current-voltage	Voltage-charge	Impedance $Z(s) = V(s)/I(s)$	Admittance $Y(s) = I(s)/V(s)$
 Capacitor	$v(t) = \frac{1}{C}\int_0^t i(\tau)d\tau$	$i(t) = C\frac{dv(t)}{dt}$	$v(t) = \frac{1}{C}q(t)$	$\frac{1}{Cs}$	Cs
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R}v(t)$	$v(t) = R\frac{dq(t)}{dt}$	R	$\frac{1}{R} = G$
 Inductor	$v(t) = L\frac{di(t)}{dt}$	$i(t) = \frac{1}{L}\int_0^t v(\tau)d\tau$	$v(t) = L\frac{d^2q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$

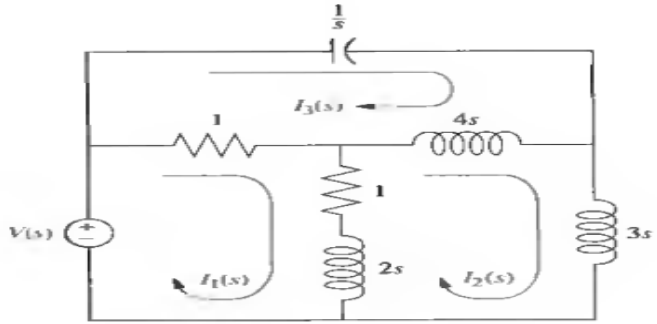
Example

1. Find the transfer function relating the capacitor voltage $V_c(s)$ to the input voltage, $V(s)$

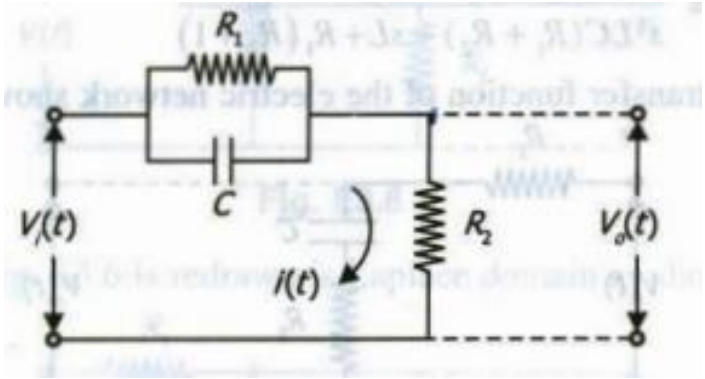


Exercise

1. Find the transfer function relating the current $I_2(s)$ to the input voltage, $V(s)$



2. Find the transfer function relating the resistor voltage $V_o(s)$ to the input voltage, $V(s)$



Modeling of Mechanical system

Mechanical system is classified into two

1. Translational Mechanical system
2. Rotational Mechanical system

1. Translational Mechanical system

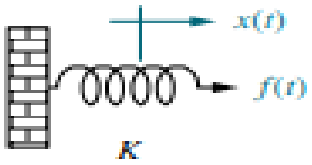
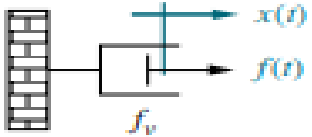
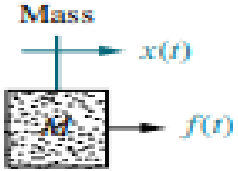
Mechanical system obey the basic law that the sum of the forces must equals to zero.

Newton's Law

'The sum of the applied force must be equal to the sum of the reactive force'

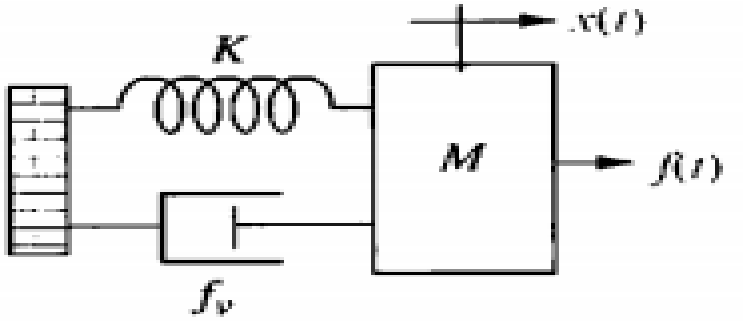
- The three quantities characterizing elements in mechanical translational system are:
  - Mass
  - Elastic
  - Damping

Modeling of Mechanical system

Component	Force-velocity	Force-displacement	Impedence $Z_M(s) = F(s)/X(s)$
Spring 	$f(t) = K\int_0^t v(\tau)d\tau$	$f(t) = Kx(t)$	K
Viscous damper 	$f(t) = f_v v(t)$	$f(t) = f_v \frac{dx(t)}{dt}$	$f_v s$
Mass 	$f(t) = M\frac{dv(t)}{dt}$	$f(t) = M\frac{d^2x(t)}{dt^2}$	Ms^2

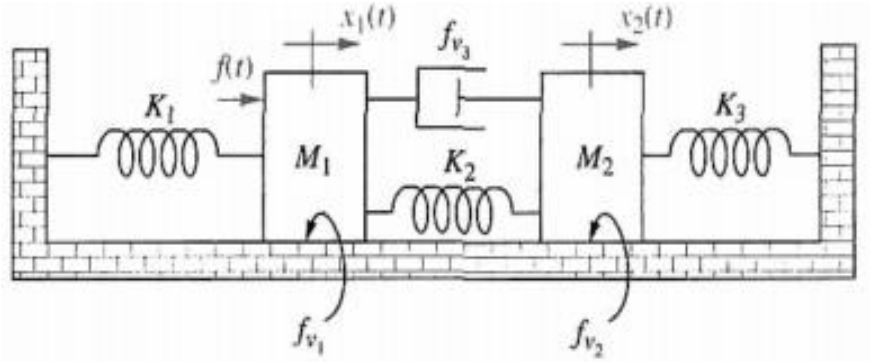
Example

1. Find the transfer function, $X(s)/F(s)$, for the mechanical system below



First draw the free body diagram. Then , make the sum of forces acting on the system is equal to zero

2. Find the transfer function, $X_2(s)/F(s)$, for the mechanical system below



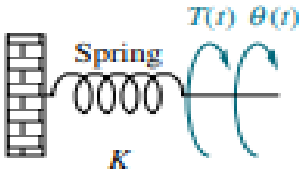
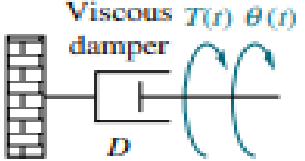
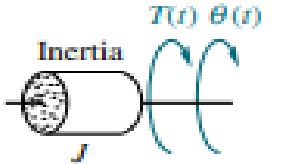
- This drawing has two masses. For the system containing two or more masses, we use **superposition principle**. i.e

1. Treat mass one fixed and the other moving.
2. Draw the free body diagram for fixed mass.
3. Treat the fixed one moving and the moving mass fixed.
4. Draw the free body diagram for the moving.
5. Draw the free body diagram sum of (2) and (4) on targeted mass.

2. Rotational Mechanical system

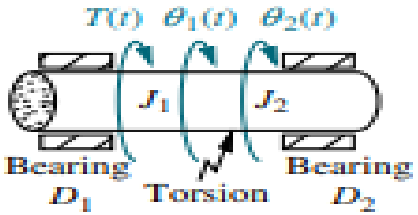
- There are many similar between translational and rotational mechanical systems

Translation	Rotational
Mass (M)	Inertia (J)
Damper (D)	Damper (D)
Spring(K)	Spring(K)
Force(F)	Torque (T)
Displacement(X)	Angular Displacement (θ)
Velocity(v)	Angular Velocity(ω)

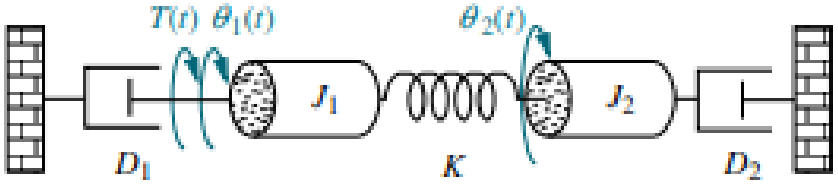
Component	Torque-angular velocity	Torque-angular displacement	Impedence $Z_M(s) = T(s)/\theta(s)$
	$T(t) = K\int_0^t \omega(\tau)d\tau$	$T(t) = K\theta(t)$	K
	$T(t) = D\omega(t)$	$T(t) = D\dfrac{d\theta(t)}{dt}$	Ds
	$T(t) = J\dfrac{d\omega(t)}{dt}$	$T(t) = J\dfrac{d^2\theta(t)}{dt^2}$	Js^2

Example

1. Find the transfer function $\theta_2(s)/T(s)$, for the rotational system shown in figure below . The rod is supported by bearings at either end and is undergoing torsion. A torque is applied at the left, and the displacement is measured at the right.



Physical system



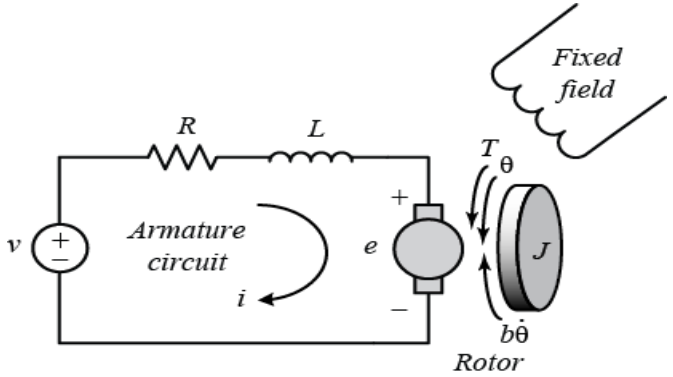
Schematic Diagram

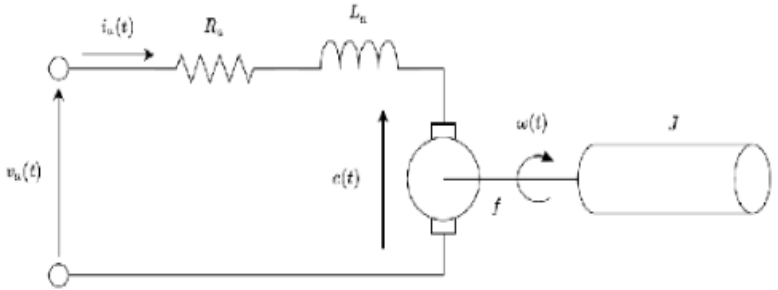
Electro Mechanical system

- Electromechanical system: are systems which have both electrical and mechanical variable.
- It has different application areas: robot control, trackers, computer tape & disk driver, different servo applications.
- **Motor:** is one of the electro mechanical system which yields a displacement output taking voltage as an input.
- For example armature controlled **DC motor:** the desired speed is obtained by varying the armature and the filed circuit but for analysis purpose, only the armature circuit considered because the filed is excited by a constant voltage.

Example

- Determine the transfer function of an armature controlled DC motor





Circuit Analogy between Electrical and Mechanical Systems:

Reading Assignment-1